Physics-Chemistry Manual

Grades 7, 8 and 9

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Chapter 1

Physics-Chemistry Manual: Grades 7, 8, and 9

1.1 Thematic Organization

This manual presents Physics-Chemistry content for grades 7, 8, and 9 organized thematically, allowing for an integrated understanding of conceptual progression in the physical and chemical sciences.

Chapter 2

PART I - PHYSICS

Chapter 3

ASTRONOMY AND GRAVITATIONAL FORCES

3.1 The Universe and the Solar System

3.1.1 Constitution of the Universe

Planets

- celestial bodies without their own light
- orbit around a star
- have sufficient mass to be approximately spherical and have no other bodies in their orbit
- can have satellites (moons) that orbit around them
- Earth is a planet

Stars

- celestial bodies with their own light
- the greater the star's mass: hotter, greater its brightness, more bluish
- in their core a chemical reaction occurs in which hydrogen is transformed into helium and much energy is released
- in their final phase they increase greatly in size and become reddish (red giants) and then contract again as: white dwarfs, neutron stars, black holes
- Earth orbits the star Sun
- other stars are only visible in the sky during the night and without light pollution

Planetary systems

- groups formed by a star and the celestial bodies that orbit around it (planets, dwarf planets, comets, asteroids, meteoroids)
- the Sun is the star of the Solar System

Star clusters

- groups of stars (that move around a point)

Galaxies

- collections of stars, gases and dust (that move around a point)
- can be spiral, elliptical or irregular
- the Solar System is located in the (spiral) galaxy Milky Way, in the Orion Arm, two-thirds of the galaxy's radius from its center

Groups and clusters of galaxies

- groups of galaxies → up to 50 galaxies
- clusters of galaxies → more than 50 galaxies
- the Milky Way is located in the Local Group

Superclusters of galaxies

- collections of groups and clusters of galaxies (that move around a point)
- the Local Group is located in the Local Supercluster (or Virgo Supercluster)

Earth's place in the Universe

Earth | Planet |

|Solar System | Planetary system | |Milky Way | Galaxy | |Local Group | Group of galaxies | |Local Supercluster | Supercluster of galaxies | |Universe | |

3.1.2 Evolution of knowledge about the Universe

Geocentrism and heliocentrism

Geocentric model

- presented by Ptolemy, in the Middle Ages
- defended that Earth was the center of the Universe and that all other celestial bodies moved around it and were immutable (with unchangeable surface)

Heliocentric model

- presented by Copernicus, in the Renaissance
- defended that the Sun was the center of the Universe and that all other celestial bodies moved around it

Galileo's discoveries

First to use a telescope

- Galileo

Galileo's observations

- the Moon is not smooth (has craters, valleys and mountains)

- Jupiter is orbited by satellites
- Venus was seen with different shapes and sizes

How these observations challenged the geocentric model

- they showed that celestial bodies were not immutable
- they showed that not all celestial bodies revolve around Earth

Catholic Church reaction

- accused Galileo of heresy, he was tried and condemned, because he challenged their ideas (that God had created Man and the World, therefore it would be the center of everything)

Currently accepted model

Current model

- the Sun is the center of the Solar System, which is just one of the planetary systems existing in the Milky Way, which is, in turn, just one of the galaxies that exist in the Universe

Space Observation

Telescopes

- magnification instruments to observe celestial bodies
- detect visible light (optical telescope) and non-visible light (radio telescope)
- can be placed on land or in orbit
  - on land: Very Large Telescope (VLC), Grande Equatorial
  - in orbit: Hubble, James Webb

Why telescopes are placed in orbit

- to not suffer interference from Earth's atmosphere and thus form sharper images

European Southern Observatory (ESO)

- responsible for the construction and use of telescopes

Why was ESO built in the desert?

- greater number of clear sky nights
- lower humidity
- less light pollution

The origin of the Universe

Big Bang Theory

- all the matter in the Universe was concentrated in a small space, at high temperature, and an explosion caused matter to expand and as it cooled the galaxies emerged

Support for the formulation of this theory

- in 1929, Edwin Hubble was able to prove that galaxies were moving away from each other, which means they were once together
- in 1964, radiation was captured by a radio telescope that was considered to be from the initial phase of the Universe

Space exploration

Space agencies

- aim to research, use technology and develop space exploration missions
- NASA (United States Space Agency)
- ESA (European Space Agency)

Missions

- manned: Vostok → took the first cosmonaut to Space, Yuri Gagarin – 1961; Apollo 11 → took 3 men to the Moon – 1969; International Space Station → since 2000
- unmanned: Sputnik 1 → first artificial satellite; Voyager 1 and 2 Probes; ExoMars Probe; Orbiter Probe

3.1.3 Distances in the Universe

Measures of distance and time

Distance units

- SI unit → meter (m)

Time units

- SI unit → second (s)

Scientific notation

Representation of numbers in scientific notation

- place in the form Y × 10x (1 Y < 10 ; x Z)
- 45500000000 = 4.5×10^1

Units for measuring distances in the Universe

Distance unit within the Solar System

- astronomical unit (au)
- corresponds to the average distance from Earth to the Sun
- 1 au = 150 million km = 1.5×10^{11} m

Distance unit outside the Solar System

- light-year (l.y.)
- corresponds to the distance traveled by light in one year
- 1 l.y. = 9.5 trillion km = 9.5×10^1 m

3.1.4 Solar System Architecture

Age of the Solar System: The Solar System formed about 4.5 billion years ago.

Constitution of the Solar System:

- **Inner Solar System:** rocky planets (Mercury, Venus, Earth, Mars)
- **Asteroid Belt:** contains most of the asteroids
- **Outer Solar System:** gas planets (Jupiter, Saturn, Uranus, Neptune)
- **Kuiper Belt:** contains asteroids and icy bodies

Main celestial bodies:

Sun:

- Only star in the Solar System
- Largest and most massive celestial body in the Solar System

Planets:

- Have translational motion around the Sun
- Orbits shaped like ellipses
- Have well-defined orbits free of debris

Dwarf planets:

- Pluto, Ceres, Haumea, Makemake, Eris
- Do not have orbital dominance

Satellites:

- Orbit exclusively planets
- Only Mercury and Venus have no satellites

Asteroids, Comets, Meteoroids:

- Asteroids: rocky bodies
- Comets: icy bodies
- Meteoroids: fragments that can form meteors (when entering the atmosphere) or meteorites (when reaching the surface)

3.1.5 Planetary Characteristics

Translation Periods:

- Increase with distance from the Sun
- Mercury: 88 Earth days (shortest period)
- Neptune: 163.72 years (longest period)

Rotation Periods:

- Vary independently of distance from the Sun

- Jupiter: 9.9 hours (shortest period)
- Venus: 243.02 days (longest period)

Satellites:

- Saturn is the planet with the most discovered satellites → 82 satellites

Rocky Planets (Terrestrial):

- Smaller
- Denser
- Hotter
- Few satellites
- Include: Mercury, Venus, Earth, Mars

Gas Planets (Jovian):

- Larger
- Less dense
- Cooler
- Many satellites
- Have rings
- Include: Jupiter, Saturn, Uranus, Neptune

Earth:

- Has one moon (natural satellite)
- Has atmosphere
- Moderate temperatures
- Only planet with conditions for life

3.2 Earth, Moon and Gravitational Forces

3.2.1 Earth's Movements

Earth's rotation movement

Earth's direction of rotation: from west to east (direct/counterclockwise direction)

Earth's rotation period: 24 hours

Consequences of Earth's rotation movement:

- succession of days and nights
- apparent motion of the Sun and stars
- variation of shadow throughout the day

How shadow varies throughout the day

Shadow direction: opposite to the Sun's

Shadow length: maximum at sunrise and sunset due to minimum Sun height and less inclined rays

Shadow length: minimum at solar noon with maximum Sun height and more inclined rays

Orientation in the northern hemisphere

During the day:

- at sunrise → shadow points west
- at solar noon → shadow points north
- at sunset → shadow points east

During the night:

- the North Star indicates north, as it remains motionless in the sky

Earth's translation movement

Earth's translation period: 365 days and 6 hours

Common year: 365 days

Leap year: 366 days (the extra day is added every four years)

Consequences of Earth's translation movement

Seasons: are due to Earth's translation movement and the inclination of Earth's rotation axis. The seasons are opposite in both hemispheres.

Solstices and equinoxes

Summer solstice: longest day of the year

Solar rays strike perpendicularly on the tropics:

- in the northern hemisphere → tropic of cancer (June 20 or 21)
- in the southern hemisphere → tropic of capricorn (December 21 or 22)

Autumn equinox: day duration equal to night, with perpendicular rays at the equator

Winter solstice: shortest day of the year

Spring equinox: day duration equal to night

Distance from Earth to the Sun

Earth is not always at the same distance from the Sun, but this difference is relatively small and is not the cause of the seasons.

Aphelion: Earth's position farthest from the Sun

Perihelion: Earth's position closest to the Sun

Seasons on other planets

Mercury: has no seasons, as its rotation axis is perpendicular to the orbital plane

Venus and Jupiter: practically have no seasons, as their rotation axis is almost perpendicular

Uranus: has its rotation axis practically lying in the orbital plane

3.2.2 Moon's Movements

Moon's rotation and translation movements

Moon's rotation period: 27 days and 8 hours

Moon's translation period: 27 days and 8 hours

Moon's orbital plane: inclined about 5º relative to Earth's orbital plane

How we see the Moon

Visible face of the Moon: is always the same due to equal rotation and translation periods

Moon phases: Full Moon, Waning Quarter, New Moon, Waxing Quarter

Why the Moon is said to be a liar: because, in the northern hemisphere, when it is waxing it has a 'D' shape and when it is waning it has a 'C' shape

Eclipses

Lunar eclipse: occurs in the Full Moon phase, with Sun-Earth-Moon alignment

Types of lunar eclipse:

- **penumbral** → the Moon is in Earth's penumbra
- **partial** → part of the Moon in the penumbra and another in the shadow
- **total** → the Moon is in Earth's shadow

Solar eclipse: occurs in the New Moon phase, with Sun-Moon-Earth alignment

Types of solar eclipse:

- **partial** → visible in the zone of Earth that is in the Moon's penumbra
- **total (or annular)** → visible in the zone of Earth that is in the Moon's shadow

Why eclipses do not occur every month: there are only eclipses when Earth, Moon and Sun are aligned

3.2.3 Forces, weight and mass

Definition of Force:

A force is a "vector quantity (represented by vectors)" characterized by value (intensity), direction, sense and point of application. The unit in SI is the newton (N), measured with a dynamometer. Results from the interaction between two bodies.

Modes of Action:

Forces act by contact or at a distance (magnetic, electric or gravitational force).

Effects of Forces:

- Deformation of a body
- Alteration of the state of rest or motion

Gravitational Force

Gravitational force is "attractive force, acting at a distance, exerted by one body on another". Its intensity depends on mass (greater mass = greater force) and distance to the center (greater distance = smaller force).

Weight corresponds to the gravitational force that a planet exerts on a body, with vertical direction, downward sense and point of application at the center of gravity.

Influencing factors on Earth:

- Greater latitude → greater gravitational force (bodies closer to the center at the poles)
- Greater altitude → smaller gravitational force

Weight and mass

Weight:

- Vector quantity
- Measured with dynamometer
- Expressed in newtons (N)
- Variable according to location

Mass:

- Scalar quantity
- Measured with balance
- Expressed in kilograms (kg)
- Constant

Mathematical Relationship:

The quantities are directly proportional. On Earth: $\mathbf{P = m \times 9.8}$ where the constant is 9.8 N/kg.

Chapter 4

MECHANICS: MOTION AND FORCES

4.1 Motion on Earth

4.1.1 Position and motion of a body

Reference Frame A **reference frame** is a coordinate system associated with a body, or bodies, that defines an origin and a set of directions, allowing the location of an object's position.

Position Position is represented by x and is measured in meters (m).

Rest vs. Motion

- **Rest:** A body is at rest when it maintains its position relative to the origin over time.
- **Motion:** A body is in motion when it varies its position relative to the origin over time.

Trajectory The **trajectory** is the set of positions occupied by a body in a given time interval. It can be classified as:

- **Rectilinear** (straight)
- **Curvilinear** (curved)

Distance Traveled Distance traveled is represented by s and is measured in meters (m).

Formula:

```
s = |xf - xi|
```

where:

- xf = final position
- xi = initial position

Time

- **Instant:** represented by t
- **Time interval:** represented by Δt

Both are measured in seconds (s).

4.1.2 Speed and velocity

Average Speed Average speed is a scalar quantity (without directional component).

Formula:

```
rm = s/Δt
```

where:

- rm = average speed
- s = distance traveled
- Δt = time interval

Velocity Velocity is a vector quantity with direction tangent to the trajectory, measured in meters per second (m/s).

Fundamental distinction:

- **Speed:** scalar quantity (magnitude only)
- **Velocity:** vector quantity (includes direction and sense)

4.1.3 Acceleration

Average Acceleration Formula:

```
am = (vf - vi)/Δt
```

where:

- am = average acceleration
- vf = final velocity
- vi = initial velocity
- Δt = time interval

Average acceleration is measured in meters per second squared (m/s^2).

Movement Types

1. **Uniform Motion:**
   - Constant velocity
   - Zero acceleration
2. **Uniformly Accelerated Motion:**
   - Increasing velocity
   - Constant positive acceleration
3. **Uniformly Retarded Motion:**

- Decreasing velocity
- Constant negative acceleration

4.1.4 Safety distance

Reaction Time Reaction time is the time interval from observing the obstacle until starting to brake.

Reaction Distance Formula:

```
dreaction = v × Δtreaction
```

where:

- dreaction = reaction distance
- v = velocity
- Δtreaction = reaction time

Braking Time Braking time is the duration of the braking process.

Braking Distance Formula:

```
dbraking = (v × Δtbraking)/2
```

where:

- dbraking = braking distance
- v = velocity
- Δtbraking = braking time

Safety Distance Total Safety Distance = Reaction Distance + Braking Distance

Factors influencing reaction time:

- Fatigue
- Alcohol
- Drowsiness
- Drugs
- Mobile phone use

Factors influencing braking:

- Weather conditions (rain, snow, ice)
- Vehicle condition (tires, suspension, brakes)
- Type of surface (dirt, asphalt)

4.1.5 Position-time graphs

Position-time graphs are fundamental tools for:

- Analyzing the motion of bodies

- Determining velocities
- Identifying types of motion
- Calculating distances traveled

The slope of the line reveals the speed: greater slope means greater speed. A horizontal line indicates rest.

4.2 Forces and motion

4.2.1 Forces

What is a force?

Force: interaction between bodies, by contact or at a distance, that can cause them deformation, as well as alter the state of rest or motion

Represented by: **F** Unit: **newton (N)** Measured with: **dynamometer**

Characterized by: value (or intensity), direction, sense and point of application

As a vector quantity.

Law of action-reaction (Newton's 3rd Law)

"To every action (exerted force) corresponds a reaction (another force) with the same intensity, the same direction, but opposite sense."

Example: A cue exerts a force on a ball and that ball exerts a force on the cue with the same intensity, the same direction, but opposite sense.

These two forces form an **action-reaction pair**.

Normal reaction

A body at rest on a flat and horizontal surface acts on that surface with a force of value equal to its weight.

Example: A book on a table exerts on it a vertical, downward force of value equal to its weight. In turn, the table exerts a vertical, upward force with the same intensity on the book.

The reaction force of the surface on the body is called: **normal reaction**

Resultant of applied forces

Resultant force (or resultant of forces): force that produces the same effects as all forces applied to a body

Calculation:

Two forces with same direction and sense: Fr=F +F

Two forces with same direction, opposite senses: Fr=F -F (if F1 > F2) Fr=F -F (if F2 > F1)

Two forces with perpendicular directions: Fr=√[(F)²+(F)²]

4.2.2 Newton's 1st and 2nd Laws

Law of Inertia (Newton's 1st Law)

When the resultant force is zero:

- if the body is at rest, it continues at rest
- if the body is in motion, it continues in uniform rectilinear motion

Justified by the **inertia** of bodies.

Inertia: property of bodies that indicates the greater or lesser difficulty in altering their state of rest or motion

It is greater the greater the mass.

Fundamental law of dynamics (Newton's 2nd Law)

When the resultant force is not zero:

- the body acquires **acceleration**

Motion results:

- If acceleration has the direction of motion: **uniformly accelerated rectilinear motion**
- If acceleration opposes the direction of motion: **uniformly retarded rectilinear motion**

For a determined resultant force, acceleration is greater when mass is smaller. Acceleration and mass are inversely proportional.

Formula:

$$Fr = m \times a$$

When the resultant force is weight: P=m×g

Where: g – gravitational acceleration

4.2.3 Forces in road safety

Collision force

The collision force is greater:

- the greater the mass
- the greater the velocity

- the smaller the collision time

Formula:

$$F_{collision} = -m \times (v_{collision}/\Delta t)$$

Pressure

Pressure: expressed as the force exerted per unit area. It is greater the smaller the area

Formula:

$$P = F/A$$

Unit: **pascal (Pa)**

Road safety elements

Safety elements increase collision time and distribute forces over larger areas to decrease pressure and reduce collision effects.

Safety elements:

- seat belt
- headrest
- airbag
- helmet
- structure with deformable zones

4.2.4 Forces resisting motion

Friction force

Friction force: force that opposes sliding or the tendency for this motion and that results from the body's interaction with the surface in contact

Increases with: surface roughness

Friction forces can be:

Useful examples:

- force that results from contact between our hand's fingers and a pen's surface to be able to hold it

Harmful examples:

- force that results from contact between the parts of a clock mechanism and that opposes motion between them

Friction can be reduced through:

- lubricants
- polishing the parts in contact

Air resistance force

Air resistance force: force that opposes the motion of bodies in air

Depends on: body shape and its velocity

Relationship: The greater the velocity the greater the air resistance

4.3 Forces, motion and energy

4.3.1 Energy transformations and transfers

Fundamental types of energy

There are two basic categories: kinetic energy and potential energy.

Kinetic energy

The kinetic energy of a body is the energy associated with the **motion** of that body.

Represents energy associated with motion. Depends on mass and velocity, increasing with both.

Formula:

$$Kineticenergy = (mass \div 2) \times (velocity)^2$$

Potential energy

Constitutes **stored** energy resulting from interactions between bodies, which can be transformed into kinetic energy.

Can be of various types:

- Elastic
- Electric
- Chemical
- Gravitational

Gravitational potential energy

Results from body-planet interaction. Depends on mass, gravitational acceleration and height.

Formula:

$$Gravitationalpotentialenergy = mass \times gravitationalacceleration \times height$$

Energy transformations

During falls, potential energy converts into kinetic energy, the sum remaining constant. Ascents reverse the process.

Fundamental principle: In the absence of air resistance, mechanical energy remains constant during the fall or rise of a body.

Mechanical energy

Corresponds to the **sum of kinetic and potential energy**. Remains constant without air resistance.

Formula:

$$Mechanicalenergy = Kineticenergy + Potentialenergy$$

Mechanical energy at specific points

At the highest point: potential energy is maximum, kinetic energy is zero.

On the ground: kinetic energy is maximum, potential energy is zero.

Energy transfer by force action

Energy transfer occurs through force actions between systems.

Work

Work is a scalar quantity that measures energy transferred by forces. It is represented by **W** and measured in joules (J).

Important note: Work requires motion to exist. Without motion, there is no work done.

Positive, negative and zero work

Work can be classified into three categories:

W $>$ 0 $\rightarrow$ positive work

- The force has the same sense as motion
- Kinetic energy increases

W $<$ 0 $\rightarrow$ negative work

- The force has opposite sense to motion
- Kinetic energy decreases

W $=$ 0 $\rightarrow$ zero work

- There is no motion
- Kinetic energy remains constant

4.4 Forces and fluids

4.4.1 Buoyancy and Archimedes' Law

Archimedes' observations in the study of bodies immersed in a liquid

"Archimedes, Greek physicist and mathematician from the 3rd century BC, studied how bodies behaved when immersed in a liquid, thus managing to explain why bodies appeared to weigh less"

The ancient investigator determined that this weight reduction resulted from an upward force exerted by the liquid. The intensity of this force did not depend on the material or weight of the body, but exclusively on the volume of liquid displaced. He also verified that the force corresponded to the weight of the displaced fluid.

"These observations were also verified in gases, although with much lower intensity, and were expressed in Archimedes' Law."

Archimedes' Law

"According to Archimedes' Law, any body immersed in a fluid undergoes from it a vertical force, from bottom to top, whose intensity is equal to the weight of the fluid displaced by the body."

Note: "fluid is any material that acquires the shape of the container that holds it (liquid or gaseous material)."

Buoyancy

"The force characterized by Archimedes is called buoyancy. We can define buoyancy as the upward force exerted by a fluid on a body that is totally or partially immersed in it, whose intensity is equal to the weight of fluid displaced by the body."

Apparent weight

"It is due to the buoyancy force that bodies appear to weigh less inside a fluid like water."

"The weight of a body immersed in a fluid is called apparent weight."

The intensity of buoyancy can be determined by the following relationship:

Buoyancy = Real weight − Apparent weight

Note: "the weight of the body in air is considered its real weight since the buoyancy that air exerts on the sphere is practically zero."

Buoyancy and motion of bodies in a fluid

"The relationship between weight and buoyancy exerted on a body determines the motion that body performs inside the fluid. When weight is greater than buoyancy the body descends, when it is smaller it rises, and when it is equal it remains in equilibrium."

Weight > Buoyancy → the body descends

Weight < Buoyancy → the body rises

Weight = Buoyancy → the body remains in equilibrium

Buoyancy and flotation of bodies

"When a body immersed in a fluid has weight less than buoyancy, it rises until the buoyancy value becomes equal to the weight value, remaining floating."

"Buoyancy decreases until it equals the body's weight due to the part of the body that remains immersed being smaller (smaller volume of immersed body → smaller volume of displaced fluid → smaller buoyancy)."

Factors influencing buoyancy

"Buoyancy is greater the greater the volume of displaced fluid and the greater the density (or volumetric mass) of the fluid."

Formulas to calculate buoyancy

"Buoyancy does not depend on the body's weight. According to Archimedes' Law, it corresponds to the weight of displaced fluid."

Buoyancy = weight of displaced fluid

"Since the weight of a body equals the product of its mass with gravitational acceleration, buoyancy can be calculated by multiplying the mass of displaced fluid by gravitational acceleration."

Buoyancy = mass of displaced fluid × gravitational acceleration

"Given that buoyancy depends on the fluid's density and the volume of displaced fluid, it is also possible to calculate buoyancy knowing their values:"

Buoyancy = density of displaced fluid × volume × gravitational acceleration

Note: "density (or volumetric mass) = mass : volume mass = density × volume"

"Finally, and as we have seen, it is also possible to calculate buoyancy through the difference between the body's real weight and its apparent weight when immersed in a fluid."

Buoyancy = weight − apparent weight

Applications of Archimedes' Law

Ships: "Ships have high weight because they are made of dense materials like iron and steel. However, they manage to float because the shape of the lower part is designed to displace a large amount of water and so that buoyancy increases when the ship's weight increases due to cargo."

Submarines: "Submarines are another example of buoyancy application. In their hull they have tanks that can be filled with water to increase weight and submerge and they empty them through compressed air to emerge."

Hot air balloons: "In air, we have the example of hot air balloons. The balloon's volume is quite high so it can displace a large amount of air and the hot air inside the balloon has less mass than

the cold air that was displaced, making buoyancy greater than the balloon's air weight, allowing it to rise."

Chapter 5

WAVES: SOUND AND LIGHT

5.1 Waves and Sound

5.1.1 Types of Waves

Transverse Waves:

- The vibration of particles is perpendicular to the direction of wave propagation

Longitudinal Waves:

- The vibration of particles is parallel to the direction of wave propagation

5.1.2 Sound and Propagation

Sound propagates through air as longitudinal waves. These waves are represented as pressure waves:

- **Crests:** represent compression zones
- **Troughs:** represent rarefaction zones

5.1.3 Wave Characteristics

Amplitude (A):

- Maximum distance relative to equilibrium position
- Measured in meters

Wavelength (- lambda):

- Minimum distance between two particles in the same vibration phase
- Distance between equivalent vibration points

Frequency (f):

- Number of vibrations per second
- Measured in Hertz (Hz)

Period (T):

- Time necessary for one complete vibration

5.1.4 Mathematical Relationships

Relationship between frequency and period:

```
f = 1/T
```

Wave velocity calculation:

```
v =  /T
```

Where:

- v = wave velocity
- = wavelength
- T = period

5.1.5 Sound production and transmission

Sound Sound is produced through the vibration of a body.

Vibration Vibration is the repetitive motion of a body, or part of it, around an equilibrium position.

Sound source A sound source is a vibrating body that produces sound. Examples include:

- Vocal cords
- Musical instruments:
  - Wind instruments
  - String instruments
  - Percussion instruments

Frequency Frequency is the number of vibrations per second, measured in hertz (Hz).

Frequency formula:

```
f = number of vibrations / Δt
```

Where:

- f = frequency
- Δt = time interval

Acoustics Acoustics is the area of Physics that studies sound.

Propagation Media Sound propagates through:

- Solids
- Liquids
- Gases

Important: Sound does NOT propagate in vacuum - it needs a material medium.

Transmission Mechanism Sound transmission occurs through mechanical waves that involve:

- **Compression zones**: regions of greater density and pressure
- **Rarefaction zones**: regions of lower density and pressure

During propagation:

- Air layers transfer energy without moving
- The source's frequency is maintained throughout propagation
- Medium particles vibrate but do not move along the medium

Sound Velocity

Velocity formula:

```
v = d / Δt
```

Where:

- v = velocity
- d = distance
- Δt = time interval

Factors affecting velocity:

1. **Propagation medium:**
   - Solids: fastest velocity
   - Liquids: intermediate velocity
   - Gases: slowest velocity
   - Order: v(solids) > v(liquids) > v(gases)
2. **Temperature:**
   - Velocity increases with temperature increase

Acoustics Applications

Acoustics science has several practical applications:

1. **Medicine**: Ultrasound for medical diagnosis
2. **Fishing**: Sonar for locating schools of fish
3. **Architecture**: Acoustic design of spaces
4. **Music**: Development and design of musical instruments
5. **Environmental protection**: Noise reduction systems

5.1.6 Sound attributes and detection by humans

Sound Pitch: Related to the frequency of sound waves. High-pitched sounds have high frequency, while low-pitched sounds have low frequency. Pitch varies according to the mass, length and characteristics of the instrument.

Sound Intensity: Associated with the amplitude of the sound wave. Loud sounds have high intensity; weak sounds, low intensity. Amplitude decreases with distance from the source.

Timbre: Attribute that allows distinguishing complex sounds with the same pitch and intensity. The tuning fork produces pure sounds, while typical musical instruments generate complex sounds with harmonics.

Human Detection

The human ear is divided into three functional parts: outer ear (captures waves), middle ear (amplifies vibrations) and inner ear (converts to nerve signals).

The human sound spectrum lies between 20 Hz and 20,000 Hz. Sounds below this are infrasound; above are ultrasound. Sound intensity is measured in decibels, with hearing threshold at 0 dB and pain threshold at 120 dB.

5.1.7 Acoustic phenomena

Absorption

"Sound absorption occurs when sound waves encounter rough surfaces, energy being dissipated"

Materials with high absorption capacity:

- Fabrics
- Styrofoam
- Cork

Reflection

"Sound reflection happens when waves encounter smooth and hard surfaces, sending them back"

Reflected sound has lower intensity due to partial energy absorption by the obstacle.

Reflection can produce two effects:

Echo:

- Occurs when the distance between sound source and obstacle exceeds 17 meters
- Result: audible repetition of emitted sound

Reverberation:

- Occurs when distance is less than 17 meters
- Result: prolongation of emitted sound

Resonance: Increase in sound intensity through successive reflections.

Refraction

"Sound refraction occurs when waves change direction when passing between different media"

The change in propagation velocity causes the directional change. Refracted sound has lower intensity than incident sound.

Applications of sound waves

In Medicine

Stethoscope:

- Allows hearing internal body sounds
- Based on the greater velocity of sound propagation in solids and the resonance phenomenon

Ultrasound:

- Creates images of internal tissues through reflection of ultrasound

In Navigation

Sonar:

- Detects bodies on the ocean floor using reflection of ultrasound

Echo in Nature

Echolocation: Dolphins, whales and bats use this mechanism to locate objects and other living beings.

Reverberation Reduction

Spaces like performance halls and cinemas have adequate reverberation times. Coatings are selected by absorption capacity to obtain the desired time.

Acoustic Insulation

To reduce noise pollution, materials with high absorption are used. In residences double-glazed windows and double walls with interior insulation are applied.

5.2 Light and Optical Phenomena

5.2.1 Light and light propagation

Light and Fundamental Differences

Sound waves differ from light waves: sound waves are longitudinal, mechanical and do not propagate in vacuum, while light waves are transverse, electromagnetic and propagate in vacuum.

Electromagnetic Waves

An electromagnetic wave results from an "electric and magnetic disturbance" characterized by frequency, period and propagation velocity. The maximum transmission velocity is the speed of light in vacuum: $\mathbf{c = 299\ 792\ 458\ m/s}$

Electromagnetic Spectrum

The spectrum includes (in order of increasing energy):

- Radio waves (communications, television)
- Microwaves (mobile phones, GPS, ovens)
- Infrared (heaters, remote controls)
- **Visible light** (illumination)
- Ultraviolet (tanning beds)
- X-rays (radiography, CT)
- Gamma rays (PET, nuclear energy)

Optical Concepts

Optics: study of light. Light is represented by "light beams" (parallel, convergent or divergent).

Bodies: are classified as luminous (emit light) or illuminated (reflect received light).

Light-Matter Interaction

When light interacts with bodies the following occurs:

- Reflection (sending back)
- Absorption
- Transmission
- Dispersion (white light $\rightarrow$ rainbow colors)

Materials:

- Transparent (form sharp images)
- Translucent (diffuse image)
- Opaque (no transmission)

Shadow and Penumbra

Since "light propagates in a straight line," opaque bodies create:

- **Shadow:** absence of visible light
- **Penumbra:** decrease in light intensity

Light Diffusion

Visualization of light rays requires suspended particles that reflect light to the eyes.

5.2.2 Optical phenomena

Light Reflection

Definition and Types:

- **Regular (specular) reflection**: Occurs on polished surfaces. A parallel beam of incident light originates a parallel beam of reflected light.
- **Irregular (diffuse) reflection**: Occurs on rough surfaces.

- Both types of reflection occur simultaneously, with one predominating over the other.

Applications:

- Mirrors
- Periscopes
- Radar

Mirrors and Images

Types of Images:

- **Real images**: Can be projected on a screen
- **Virtual images**: Only visible to the eyes, cannot be projected

Flat Mirrors: Characteristics of formed images:

- Virtual image
- Same size as object
- Upright (not inverted vertically)
- Symmetric relative to the mirror

Curved Mirrors:

- **Convex mirrors**: Produce virtual images smaller than the object
- **Concave mirrors**: Can produce real or virtual images depending on object position

Light Refraction

Definition: "Alteration of light velocity when changing propagation medium"

Refringence: Property of material media related to light propagation velocity.

Relative Velocities: Light velocity varies in different propagation media. Light velocity in glass depends on light frequency.

Explanation: When light passes from one medium to another, alteration in its velocity occurs, which causes the refraction phenomenon. Different colors (frequencies) deviate differently, explaining phenomena like dispersion.

Lenses

Types of Lenses:

Convergent (Convex) Lenses:

- Produce real images
- Inverted images

Divergent (Concave) Lenses:

- Produce virtual images
- Upright images (not inverted)

- Images smaller than object

Focal Power: Calculation formula:

P = 1/f

Where:

- P = focal power (measured in diopters)
- f = focal distance

The unit of measurement is the diopter.

Human Vision

Eye Function: The eye functions as a light receptor. The crystalline lens (eye lens) forms inverted images on the retina.

Vision Defects:

Myopia:

- Corrected with divergent lenses

Hyperopia:

- Corrected with convergent lenses

Color and Light Dispersion

Monochromatic vs. Polychromatic Light:

- Monochromatic light: A single frequency/color
- Polychromatic light: Multiple frequencies/colors

Primary Colors: Three primary colors:

- Red
- Green
- Blue These three colors combine to create all other colors.

Light Dispersion: Phenomenon that explains rainbow formation. When white (polychromatic) light passes through a medium with different refractive indices for different frequencies, the colors separate.

Color of Objects: The color of an object depends on the components of incident light that are reflected. Objects appear colored based on the light they reflect from the incident illumination.

Chapter 6

ELECTRICITY

6.1 Electric Current and Circuits

6.1.1 Electric current

What is electric current?

- **Electric current**: oriented movement of particles with electric charge (electrons or ions)

Good and bad electrical conductors

- **Good electrical conductors**:
  - have free electric charges to move
  - examples: metals, graphite and aqueous solutions
- **Bad electrical conductors**:
  - do not have free electric charges to move
  - examples: wood, cork, rubber, plastic, distilled water

6.1.2 Electrical circuits

What is an electrical circuit?

- **Electrical circuit**: sequence of components through which electric current can flow

Components of an electrical circuit and their representation

- **Energy source (supplies energy to the circuit)**:
  - battery
  - generator
  - electrical outlet
- **Receivers (transfer or transform electrical energy)**:
  - lamp

  - resistor
  - rheostat
  - motor

- **Connecting wires** (connect the various circuit components)
- **Switches** (open or close the circuit):
  - open switch: opens the circuit
  - closed switch: closes the circuit
- **Measuring devices**:
  - voltmeter
  - ammeter

Series and parallel circuits

- **Series circuit**: electric current flows through a single circuit
- **Parallel circuit**: electric current flows through the main circuit and branches

6.1.3 Electric current quantity

Characteristics of electric current

- **Electric current (I)**:
  - relates the quantity of electric charges that crosses a cross-section of a conductor per unit of time
  - has as SI unit the **ampere (A)**
  - is measured with an **ammeter** which is installed in series

Multiples and submultiples of the ampere unit

- kA – kiloampere
- mA – milliampere

1 kA = 1000 A

1 mA = 0.001 A

Direction of electric current

- **Real direction**: from negative pole to positive pole
- **Conventional direction**: from positive pole to negative pole

Direct current and alternating current

- **Direct (DC)**: maintains direction
- **Alternating (AC)**: varies direction

Measuring electric current

- **With lamps in series**:

I = I = I = …

- **With lamps in parallel**:

I = I + I + …

6.1.4 Voltage quantity

Characteristics of voltage (or potential difference)

- **Electric voltage (U)**:
  - corresponds to the energy that a generator supplies to the electrical circuit per unit of charge
  - has as SI unit the **volt (V)**
  - is measured with a **voltmeter** which is installed in parallel

Multiples and submultiples of the volt unit

- kV – kilovolt
- mV – millivolt

1 kV = 1000 V

1 mV = 0.001 V

Volta's Battery

Volta's battery was the first electrochemical generator. It consisted of copper and zinc disks, interspersed with cloth moistened with saline solution.

Series association of batteries

The voltage at the poles of a series association of batteries equals the sum of the voltage at the poles of each battery:

U(batteries in series) = U(battery 1) + U(battery 2) + …

Measuring electric voltage

- **With lamps in series**:

U = U + U + …

- **With lamps in parallel**:

U = U = U = …

6.1.5 Electrical resistance quantity

Characteristics of electrical resistance

- **Electrical resistance (R)**:
  - measures the opposition that a conductor offers to the passage of electric current
  - has as SI unit the **ohm (Ω)**
  - is measured with an **ohmmeter**

Multiples and submultiples of the ohm unit

- kΩ – kilo-ohm
- mΩ – milliohm

1 kΩ = 1000 Ω

1 mΩ = 0.001 Ω

Measuring electrical resistance

We can measure a conductor's resistance directly through an ohmmeter, or indirectly through a voltmeter and ammeter, since for constant voltage, electric current is inversely proportional to the conductor's resistance:

R = U / I

Ohm's Law

At constant temperature, the quotient between the voltage at the terminals of an ohmic conductor and the electric current flowing through it is constant.

Factors on which electrical resistance depends

Electrical resistance is greater:

- the longer the conductor
- the smaller the cross-sectional area of the conductor

Rheostat

A **rheostat** is an electrical component with variable resistance.

6.2 Effects of electric current and electrical energy

6.2.1 Electrical power of a receiver

Characteristics of electrical power

- **Electrical power (P):** energy transferred per unit of time; has as SI unit the watt (W)
- **Energy (E):** has as SI unit the joule (J)

Other units for power and energy

- **Multiple of the watt unit:** kW – kilowatt; 1 kW = 1000 W
- **Unit commonly used for energy:** kW h – kilowatt-hour; 1 kW h = 3,600,000 J

Calculating electrical power

- $P = E/\Delta t$
- $P = U \times I$

Nominal values of an electrical receiver

"Electrical receivers have characteristic power values and voltage suitable for their operation. These are called nominal values."

6.2.2 Electrical energy in homes

Power to contract for a home

Depends on:

- the electrical power of each appliance
- the number of electrical appliances

Phase and neutral

"The electrical outlets where appliances are plugged in have two terminals:"

- the **phase**: which varies voltage
- the **neutral**: which does not vary voltage

"The voltage variation of the phase makes it have a higher or lower value than the neutral, which makes the current alternating."

Short circuit

- **Short circuit:** sudden increase in electric current, usually caused by direct contact between generator poles (phase and neutral)

Safety rules and short circuit prevention:

- use fuses (have a metallic wire that melts easily when there is high electric current, thus interrupting the circuit)
- use circuit breakers (interrupt the electrical circuit by magnetic effect when electric current values are very high)
- protect electrical outlets
- do not use or clean appliances with wet hands

6.2.3 Effects of electric current

Thermal effect (Joule effect)

- **Thermal effect:** release of energy as heat; due to heating of metallic conductors due to the resistance they offer to the passage of electric current
- **Applications:** heaters

Chemical effect

- **Chemical effect:** occurrence of chemical reactions due to the passage of electric current; example: electrolysis of water which decomposes water into dihydrogen and dioxygen
- **Applications:** metal coating

Magnetic effect

- **Magnetic effect:** induction of a magnetic field
- **Applications:** electric motors, speakers, electromagnets

Effects of electric current on living beings

"The human body is an electrical conductor and, therefore, is susceptible to receiving electric shocks. It can receive a non-painful shock, but if the electric current is high it can cause painful muscle contractions, burns, cardiac arrhythmia and even death."

Chapter 7

ENERGY

7.1 Sources and Energy Transfers

7.1.1 Energy and energy sources

Energy associated with systems

- **Physical system**: body or set of components subject to a physical study
- **Internal energy**: energy of a system
- **Types of systems**:
  - **Isolated system**: does not allow exchanges of energy or matter with the exterior
  - **Closed system**: allows exchanges of energy but not matter with the exterior
  - **Open system**: allows exchanges of energy and matter with the exterior

How energy is measured

- **SI unit of energy**: joule (J)
- **Commonly used unit**: calorie (cal) - 1 cal = 4.18 J

Energy sources in nature (primary)

- **Renewable energy sources**:
  - Advantages: do not run out; low polluting
  - Disadvantages: high investment cost; negative impact on some ecosystems
  - Examples: sun, geothermal, wind, biomass, water
- **Non-renewable energy sources**:
  - Advantages: economic profitability; do not depend on atmospheric conditions
  - Disadvantages: exhaustible; polluting; environmental disasters with oil tankers; risks of nuclear accidents
  - Examples: coal, oil, natural gas, uranium

Secondary energy sources

- **Electricity**: obtained from power plants (hydroelectric, thermal, nuclear, wind, solar)

- **Gasoline, diesel and butane gas**: obtained from oil

7.1.2 Energy transfers

Energy transfer and energy conservation

- **Energy transfer**: passage of energy from one system (source) to another (receiver)
- **Energy conservation**: energy is neither created nor destroyed, only transferred or transformed; the total energy of the Universe is constant

Thermal energy

- **Temperature**: quantity that relates to the agitation of corpuscles in a body. Higher temperature implies greater agitation and greater internal energy. Measured in degrees Celsius (°C) with thermometers.

Energy transfer as heat

- **Heat**: energy transfer that occurs spontaneously between bodies in contact with different temperatures until they reach thermal equilibrium. Occurs from the body of higher temperature to that of lower temperature.

Energy transfer processes

- **Thermal conduction**: occurs with contact between bodies (essentially solids); energy transfers particle to particle; no movement of matter; good thermal conductors (metals) transfer more quickly than poor conductors (wood)
- **Thermal convection**: occurs with heating of fluids (gases and liquids); corpuscles move creating convection currents
- **Radiation**: energy transfers through light propagation (electromagnetic waves); does not need medium; does not require contact between bodies

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Chapter 8

PART II - CHEMISTRY

Chapter 9

CONSTITUTION AND PROPERTIES OF MATTER

9.1 Materials and Laboratory

9.1.1 Constitution of the Material World

Chemistry

Chemistry is defined as "the science that studies materials and their transformations." Its objectives include producing new materials useful for society and improving the properties of some existing materials.

Classification of Materials by Origin

Materials are classified as:

- **Organic**: have animal or plant origin (milk, eggs, wool, silk, linen, cotton)
- **Inorganic** (or minerals): extracted from the Earth's crust (iron, lime, granite, quartz, salt)

Classification by Provenance

- **Natural**: found in nature
- **Natural manufactured**: natural materials transformed by humans
- **Synthetic**: manufactured in laboratories or industry (nylon, varnish, paint, plastic, medicines)

Renewable and Non-Renewable Natural Resources

Renewable resources have a rapid replacement rate (water, soil, forest). Non-renewable resources have slow replacement: fossil fuels (oil, coal, natural gas), metallic mineral resources (aluminum, copper, gold, zinc, iron) and non-metallic (limestone, marble, granite).

Rational Use of Resources

All resources are limited. Waste in landfills is not reused. It is recommended to refuse single-use products, reduce, reuse, recover and recycle. Composting transforms organic materials into natural fertilizer.

9.1.2 Substances and Mixtures

Fundamental Difference

"Substances" consist of a single component (oxygen, distilled water, sodium chloride). "Mixtures" have two or more components (air, mineral water, seawater, steel, brass).

Classification of Mixtures

- **Homogeneous**: components cannot be distinguished (air, brass, water with dissolved sugar)
- **Heterogeneous**: components can be distinguished (granite, water with oil)
- **Colloidal**: appear homogeneous but show heterogeneous under microscope (blood, ink, mayonnaise)

Miscible and Immiscible Liquids

Miscible liquids form homogeneous mixture (water with ethyl alcohol). Immiscible liquids form heterogeneous (water with oil).

9.1.3 Solutions

Definition and Components

A solution is a "homogeneous mixture (solid, liquid or gaseous)." Includes solute(s) - substance(s) that dissolve(s) - and solvent - substance that dissolves.

Types of Solutions

Aqueous solutions have water as solvent. Alcoholic solutions use alcohol.

Composition

Qualitative composition identifies solute(s) and solvent. Quantitative composition indicates quantity of each component.

Concentration

More concentrated solutions have greater quantity of solute. Diluted have lesser quantity. Concentration is calculated: Cm = (mass of solute)/(volume of solution). The SI unit is Kg/m^3; in laboratory g/cm^3 is used.

Solubility

A saturated solution contains the maximum amount of solute that the solvent can dissolve at a given temperature.

9.2 Properties of Materials

9.2.1 Physical Properties

Melting and Boiling Points:

- "Melting point" is defined as "temperature at which a substance passes from solid state to liquid state, at a given pressure"
- "Boiling point" is "temperature at which a substance passes from liquid state to gaseous state, at a given pressure"
- For water: melting point is 0°C and boiling point is 100°C (at normal atmospheric pressure)

Density/Mass Volume:

- Formula: "density ( ) = mass (m)/volume (v)"
- Units: Kg/m³ (SI) or g/cm³ (laboratory)
- Water density: 1 g/cm³ (at 4°C)

Special note about ice:

- Ice maintains mass but increases volume when water freezes
- This is an important exception in water behavior

9.2.2 Chemical Properties

Test for Water:

- Water turns blue anhydrous copper sulfate

Test for Starch:

- Starch turns blue-purple iodine water

Test for Carbon Dioxide:

- Carbon dioxide turns limewater cloudy

9.2.3 Physical and chemical transformations

What are physical transformations

- Physical transformations: changes in the physical form of materials without formation of new substances
- Dissolution of a substance
- Deformation of a body
- Change in degree of division
- Change in physical state

Changes in physical state

With temperature increase (energy absorption):

- Fusion: solid → liquid
- Vaporization: liquid → gaseous
- Sublimation: solid → gaseous

With temperature decrease (energy release):

- Condensation: gaseous → liquid
- Solidification: liquid → solid or gaseous → solid

Water cycle

Processes involved in the water cycle:

- Evaporation of water from rivers, lakes, seas and oceans
- Transpiration of living beings
- Condensation (cloud formation), solidification and fusion of water
- Precipitation of water
- Infiltration and runoff of water returning it to seas and oceans

What are chemical transformations

- Chemical transformations: transformations in materials in which new substances are formed
- Reactants: initial substances
- Reaction products: new substances

How to detect the occurrence of a chemical reaction

- Formation of a solid
- Formation of a gas
- Color change
- Formation of flame

What causes the occurrence of a chemical reaction

- Junction of substances
- Heating
- Action of electricity
- Action of light
- Mechanical action (movement)

How to represent a chemical reaction

- Word equation: reactants → reaction products

How to read a word equation

- "+" in reactants reads as "reacts with"
- "→" reads as "produces" or "producing"
- "+" in products reads as "and"

- Example: "dihydrogen (g) + dioxygen (g) → water (l)" reads as "gaseous dihydrogen reacts with gaseous dioxygen producing liquid water"

What are decompositions

- Decompositions: chemical reactions in which a single substance transforms into other less complex ones
- Photolysis → by action of light
- Thermolysis → by action of temperature
- Electrolysis → by action of electricity

Example of water electrolysis

- Word equation: water (l) → dihydrogen (g) + dioxygen (g)
- Performed in a voltameter consisting of two electrodes fixed at the bottom of a tub where water is placed
- In the tube placed over the negative electrode, dihydrogen is released
- In the tube placed over the positive electrode, oxygen is released
- The volume of dihydrogen is double the volume of dioxygen

What are syntheses

- Syntheses: chemical reactions in which two or more substances produce a single product
- Example: dihydrogen (g) + dinitrogen (g) → ammonia (g)

Importance of syntheses

- Production of new and better materials in a more economical and ecological way
- Ammonia, medicines, plastics, …

9.2.4 Separation of substances in a mixture

Physical processes for separating components of a mixture

Separation of components of heterogeneous mixtures

- **Magnetic separation (magnet)**
  - magnets separate solids with magnetic properties
  - iron filings + sand
- **Sieving (sieve)**
  - sieve retains particles of smaller dimensions
  - corn + flour
- **Filtration (filter)**
  - filter retains particles suspended in a liquid
  - bran + water

- **Centrifugation (centrifuge)**
  - solid in suspension deposits at the bottom of the test tube
  - ash + water
- **Solid-liquid decantation (rod)**
  - deposited solid remains at bottom and liquid is poured into another container through the rod
  - sand + water
- **Liquid-liquid decantation (separatory funnel)**
  - denser liquid is the first to be decanted
  - oil + water

Separation of components of homogeneous mixtures

- **Crystallization**
  - the most volatile substance evaporates slowly
  - sodium chloride + water
- **Distillation (distillation flask and condenser)**
  - separation of substances according to their volatility
  - water + alcohol

Physical-chemical treatment of water

Water treatment stations

- **WTP (Water Treatment Plant)**
  - makes water suitable for consumption
- **WWTP (Wastewater Treatment Plant)**
  - treats water used by humans (in homes, industries, etc.) before being returned to nature
- **Processes used in water treatment stations**
  - Physical processes:
    - screening
    - decantation
    - filtration
  - Chemical processes:
    - disinfection

Chapter 10

CHEMICAL REACTIONS

10.1 Representation and Types of Reactions

10.1.1 Corpuscular nature of matter

Corpuscular theory: "All substances are constituted by tiny corpuscles that are in constant motion, between which there is empty space."

Types of corpuscles: atoms, molecules and ions.

Physical States and Movement of Corpuscles

Solid state: corpuscles close together and organized, strong bonding forces, little freedom of movement.

Liquid state: corpuscles more distant, weaker bonding forces, some freedom of movement.

Gaseous state: corpuscles distant and disorganized, very weak bonding forces, great freedom of movement.

Temperature: "The higher the temperature, the greater the agitation of the corpuscles."

Shape and Volume

Solids have defined shape and volume. Liquids adapt to container shape but maintain volume. Gases adapt shape and volume to container.

Pressure of a Gas

"The pressure of a gas corresponds to the intensity of the force that corpuscles exert, per unit of area, on the surface of the container."

Variations:

- Same temperature and greater volume = lower pressure
- Same volume and higher temperature = higher pressure

10.1.2 Atoms

Constitution: nucleus (protons + neutrons) and electron cloud (electrons).

Electric charges: protons (positive), neutrons (neutral), electrons (negative).

Atom charge: neutral, because protons = electrons.

Mass: concentrated in nucleus; electrons have negligible mass.

Chemical element: atoms of the same type (same number of protons) represented by universal chemical symbols.

Examples of elements: H (hydrogen), C (carbon), O (oxygen), N (nitrogen), Na (sodium), K (potassium), Ca (calcium), Mg (magnesium), Al (aluminum), Cl (chlorine), S (sulfur).

10.1.3 Molecules

Definition: "Set of atoms interconnected, structural unit of substances, electrically neutral, represented by a chemical formula."

Qualitative and quantitative composition: describes which atoms and how many.

Examples:

O (dioxygen): 2 oxygen atoms (elementary substance).

H O (water): 2 hydrogen atoms + 1 oxygen (compound substance).

CO (carbon dioxide): 1 carbon + 2 oxygens per molecule.

Elementary substance: constituted by atoms of the same element.

Compound substance: constituted by atoms of different elements.

10.1.4 Ions

Negative ion (anion): atom or group that captured electrons.

Positive ion (cation): atom or group that ceded electrons.

Representation: chemical formula with symbols and charge (e.g., Cl , Ca^2).

Monoatomic ions: a single atom (Na , Cl).

Polyatomic ions: group of atoms (NH , SO^2 , NO , CO^2 , PO^3 , OH , O^2).

Ionic substances: junction of anions and cations, electrically null (equal charges).

Writing formulas: positive ion placed first, then negative, balancing charges. Example: Ca^2 + 2Cl $\rightarrow$ CaCl .

10.1.5 Representation of chemical reactions

Law of Conservation of Mass

"In a chemical reaction there are rearrangements of the atoms of the reactants that lead to the formation of new substances, conserving the total number of atoms of each element."

Lavoisier's Law: "In a chemical reaction there is no mass variation, that is, the total mass of reactants equals the total mass of reaction products."

Word and Chemical Equations

Example - Combustion of hydrogen:

Word equation: Oxygen + Hydrogen → Water

Initial chemical equation: $O + H \rightarrow HO$ (unbalanced)

Balanced equation: $2H + O \rightarrow 2HO$

Explanation: 2 hydrogen molecules + 1 oxygen molecule = 2 water molecules, conserving 4 H atoms and 2 O atoms on both sides.

10.1.6 Types of chemical reactions

Oxidation-reduction reactions

Types of oxidation-reduction reactions

Combustion reactions:

- Reactions between a fuel (substance that burns) and an oxidizer (substance that enters combustion)
- Release of large quantities of energy
- Generally involve appearance of flame and, if very fast, can occur with explosion
- Require activation energy (heat, flame or electric discharge)

Cellular respiration:

- Energy release
- Involves nutrients that ensure cell functioning

Oxidation of metals (corrosion):

- Release of some energy
- Slow process that transforms metals into metal oxides

Emission of polluting gases

The use of fossil fuels produces a wide variety of polluting gases:

- Sulfur dioxide (SO2)
- Nitrogen oxides (NO, NO2)

- Carbon dioxide (CO2)

Effects:

- Acid rain (SO2, NO, NO2)
- Depletion of ozone layer (NO, NO2)
- Greenhouse effect (CO2)

Solutions:

- Addition of substances during combustion
- Use of emission filters

Acid-base reactions

Acidic, basic and neutral solutions

Acidic solutions:

- Have sour taste
- Corrode some metals
- Some can be aggressive to living tissue
- Examples:
  - Sulfuric acid (H2SO4)
  - Nitric acid (HNO3)
  - Hydrochloric acid (HCl)
  - Phosphoric acid (H3PO4)

Basic solutions (alkaline):

- Have bitter taste
- Are slippery to touch
- Some can be aggressive to living tissue
- Examples:
  - Ammonia (NH3)
  - Calcium hydroxide (Ca(OH)2)
  - Sodium hydroxide (NaOH)
  - Potassium hydroxide (KOH)
  - Magnesium hydroxide (Mg(OH)2)
  - Sodium hydrogen carbonate (HCl)

Neutral solutions:

- Examples:
  - Distilled water
  - Sodium chloride

pH scale (Sorensen scale)

Acidic solutions:

- Have pH between 0 and 7
- The lower the pH value the greater the acidity

Basic solutions (alkaline):

- Have pH between 7 and 14
- The higher the pH value the greater the basicity

Neutral solutions:

- Have pH equal to 7

Indicators of chemical character of a solution

Alcoholic phenolphthalein solution (colorless):

- Turns carmine pink in basic solutions

Blue litmus solution:

- Turns red in acidic solutions
- Turns purple in neutral solutions

Blue litmus paper:

- Turns red in acidic solutions

Red litmus paper:

- Turns blue in basic solutions

Universal indicator (in solution or on paper):

- Indicates pH value according to the color it presents

Products of an acid-base reaction

The reaction between an acid and a basic substance produces:

- Salt
- Water

Name of salts:

- Hydrochloric acid gives rise to chloride
- Sulfuric acid gives rise to sulfate
- Nitric acid gives rise to nitrate
- Phosphoric acid gives rise to phosphate

Neutralization

Reactions between acids and bases are called neutralization reactions as the pH value tends toward 7:

When we add drops of basic solution to an acidic solution:

- Acidity decreases
- pH value increases

When we add drops of acidic solution to a basic solution:

- Basicity decreases
- pH value decreases

Precipitation reactions

What are they?

Precipitation reactions:

- Reactions in which a precipitate (solid) is formed

Solubility of salts:

Poorly soluble salts (form precipitates):

- Hydroxides
- Carbonates
- Phosphates
- Sulfides
- Magnesium, calcium, lead and barium fluoride
- Silver, mercury, lead chlorides
- Silver, mercury, lead bromides
- Silver, mercury, lead iodides
- Barium, mercury, lead sulfates

Very soluble salts (form aqueous solutions)

Water hardness

Depending on salt concentration, waters can be classified as follows:

Soft water:

- Has high concentration of calcium and magnesium salts

Hard water:

- Has low concentration of calcium and magnesium salts
- Hinders the action of soaps (forms foam) and causes calcium deposit in pipes, washing machines, irons…

Types of hardness:

Temporary hardness:

- Has calcium and magnesium carbonates and hydrogen carbonates

- Treated by boiling water (ebullition) as it precipitates salts

Permanent hardness:

- Has calcium and magnesium chlorides, sulfates and nitrates
- Treated through anti-limescale agents

10.2 Chemical Kinetics

10.2.1 Speed of chemical reactions

How to Measure Reaction Speed

The speed of chemical reactions depends on the number of effective collisions between reactant corpuscles.

Calculation Formulas:

$$-v = quantityofreactantconsumed/transformationtime$$

- v = quantity of product formed / transformation time

Factors that Influence Reaction Speed

1. Reactant Concentration Higher concentrations increase speed because they raise the number of molecular collisions.

2. Temperature Higher temperatures accelerate reactions, causing greater agitation of corpuscles.

3. Light Provides energy that accelerates certain chemical reactions.

4. State of Division of Reactants Greater division increases contact surface, accelerating the reaction.

Catalysts and Inhibitors

Substances that participate in chemical reactions without being consumed.

- **Catalysts**: increase reaction speed
- **Inhibitors** (antioxidants and preservatives): decrease reaction speed

Collision Theory

The speed of chemical reactions depends on the number of effective collisions between reactant corpuscles.

Chapter 11

ATOMIC STRUCTURE AND PERIODIC TABLE

11.1 Atoms and Elements

11.1.1 Evolution of knowledge of atoms

Constitution of an atom:

The atom consists of 2 regions:

- **Nucleus:** contains protons (positive charge) and neutrons (no charge)
- **Electron cloud:** contains electrons (negative charge)

Protons, neutrons and electrons are the subatomic particles constituting the atom. Protons and neutrons, being in the nucleus, are called nucleons.

Electric charge of an atom:

- Nuclear charge (nuclear charge): positive
- Electron cloud charge: negative
- The atom is electrically neutral (protons = electrons)

First atomists:

Democritus proposed the existence of indivisible atoms. For the first atomists, matter consisted of atoms of different sizes and shapes according to the material.

Evolution of atomic models:

1. **Dalton's model:** indivisible atoms of different mass and size
2. **Thomson's model (plum pudding):** atoms with negative corpuscles on positive material
3. **Rutherford's model (1st planetary model):** small nucleus with orbiting electrons
4. **Bohr's model:** electrons in specific orbits with defined energy levels

5. **Electron cloud model (current):** electrons in orbitals (probability regions); the smaller the distance to nucleus, the greater the probability of finding electrons; atomic radius = distance from nucleus to electron cloud limit

11.1.2 Atoms and chemical elements

Characterization of chemical elements:

It is the number of protons that indicates the chemical element. Atoms with equal number of protons are of the same chemical element.

Chemical symbols: H (Hydrogen), He (Helium), Li (Lithium), Be (Beryllium), B (Boron), C (Carbon), N (Nitrogen), O (Oxygen), F (Fluorine), Ne (Neon), Na (Sodium), Mg (Magnesium), Al (Aluminum), Si (Silicon), P (Phosphorus), S (Sulfur), Cl (Chlorine), Ar (Argon), K (Potassium), Ca (Calcium)

Mass of subatomic particles:

The mass of an electron is negligible compared to a proton or neutron. The atom's mass is concentrated almost entirely in the nucleus.

Atomic number and mass number:

- **Atomic number (Z):** number of protons
- **Mass number (A):** number of nucleons (protons + neutrons)

Symbolic representation of an atom:

X

Where: Z = atomic number, A = mass number, X = chemical symbol

Characterization of an atom:

To characterize an atom one must indicate:

- Chemical element
- Atomic number (Z)
- Mass number (A)
- Number of protons (p)
- Number of neutrons (n)
- Number of electrons (e)
- Nuclear charge (CN): positive, equal to number of protons
- Electron cloud charge (CNE): negative, equal to number of electrons

Example: ^{23}**Na**

- Element: sodium
- Z = 11
- A = 23

- p = 11
- n = 12
- e = 11
- CN = +11
- CNE = -11

Isotopes, isotopic abundance and relative atomic mass:

- **Isotopes:** atoms with same atomic number but different mass (equal protons, different neutrons)
- **Isotopic abundance (AIN):** percentage of each isotope in nature
- **Relative atomic mass (Ar(X)):** weighted average of atomic masses of isotopes, considering their abundances

Example of oxygen:

Isotopic abundance:

- ^{1} O: 99.76%
- ^{1} O: 0.04%
- ^{1} O: 0.20%

Ar(O) = (16×99.76 + 17×0.04 + 18×0.20)/100 = 16.0044

11.1.3 Ions

What are ions:

Atoms or groups of atoms that cede or capture electrons, acquiring electric charge:

- **Positive ion (cation):** ceded one or more electrons (more protons than electrons)
- **Negative ion (anion):** captured one or more electrons (more electrons than protons)

Symbolic representation of an ion:

X or X

Where: Z = atomic number, A = mass number, X = chemical symbol, n = electrons ceded or captured

Example: ^{23}Na

- Element: sodium
- Z = 11
- A = 23
- p = 11
- n = 12
- e = 10
- CN = +11

- CNE = -10
- Ion charge = +1

11.1.4 Electronic distributions

Energy of electrons in atoms:

Electrons distribute obeying rules:

- Distribute by energy levels with well-defined value
- The farther from nucleus, the greater the energy
- Can move up levels (absorbing radiation) or down (emitting radiation)

Electrons of last level (highest energy) = valence electrons; others = core electrons.

Maximum number of electrons of each level:

Rule: $2n^2$ (where n is the level number)

- 1st level: 2 electrons
- 2nd level: 8 electrons
- 3rd level: 18 electrons

The last level (except first) maximum 8 electrons.

Electronic distributions:

Minimum energy principle: electrons preferentially occupy levels with lower energy.

Examples (Z 20):

- 1H – 1
- 2He – 2
- 3Li – 2,1
- 4Be – 2,2
- 5B – 2,3
- 6C – 2,4
- 7N – 2,5
- 8O – 2,6
- 9F – 2,7
- 10Ne – 2,8
- 11Na – 2,8,1
- 12Mg – 2,8,2
- 13Al – 2,8,3
- 14Si – 2,8,4
- 15P – 2,8,5
- 16S – 2,8,6
- 17Cl – 2,8,7

- 18Ar – 2,8,8
- 19K – 2,8,8,1
- 20Ca – 2,8,8,2

K and Ca begin filling level 4 with 8 electrons in level 3 (last level maximum 8).

Stability of electronic distributions:

Chemical reactions involve only valence electrons:

- Few valence electrons: cede electrons
- Close to 8 valence electrons: capture electrons
- 8 valence electrons: little tendency to react (noble gases)

Exceptions:

- Hydrogen: 1 electron, tends to capture 1 to complete last level
- Helium: 2 electrons, little tendency to react (last level filled)

Atoms with last level filled (noble gases) are stable. Particles with same electronic distribution = **isoelectronic**.

Example:

20Ca – 2,8,8,2

Calcium cedes 2 valence electrons:

- Resulting distribution: 2,8,8 (equal to Argon)
- Representation: Ca^2
- Ca^2 and Ar = isoelectronic particles

11.1.5 Properties of materials and Periodic Table

Evolution of periodic classification

Döbereiner (Triads) Organized elements in groups of three with similar properties, classifying them by atomic mass.

Newlands (Law of Octaves) "organized elements by their atomic mass and verified that their properties repeated every eight elements"

Mendeleev Created the Periodic Table organized by groups (columns) with similar chemical properties, ordered by atomic mass.

Moseley (Reorganization) Reorganized the table by atomic number instead of atomic mass, establishing that chemical properties depend on atomic number.

Organization of current Periodic Table

Elements are distributed by:

- **18 Groups (vertical columns)**: elements with similar chemical properties and same number of valence electrons
- **7 Periods (horizontal lines)**: elements with same number of energy levels

Location of Elements (Z 20)

Number of valence electrons → indicates the **group Number of energy levels** → indicates the **period**

Example: Oxygen (8O = 2,6)

- 6 valence electrons = Group 16
- 2 energy levels = Period 2

Classification of elements

Natural vs. Synthetic Elements There are elements found naturally and others produced artificially.

Elements in Human Body

Highest proportion:

- Oxygen
- Carbon
- Hydrogen
- Nitrogen

Lower proportion but essential: Calcium, phosphorus, potassium, sulfur, sodium, chlorine, magnesium

Properties of metals and non-metals

Metals

- Good thermal and electrical conductors
- Malleable, hard, shiny
- High melting and boiling points
- Tend to cede electrons
- Form basic/alkaline oxides

Non-Metals

- Poor thermal and electrical conductors
- Brittle (if solid)
- Volatile (if liquid)
- Tend to capture electrons
- Form acidic oxides

Families of elements

Group 1: Alkali Metals Examples: lithium, sodium, potassium

- 1 valence electron
- Form monopositive ions
- Very reactive (reactivity increases with atomic number)
- React with halogens forming salts
- *Note*: hydrogen is in group 1 but is a non-metal

Group 2: Alkaline Earth Metals Examples: magnesium, calcium

- 2 valence electrons
- Form dipositive ions
- Reactive (reactivity increases with atomic number)
- React with halogens forming salts

Group 17: Halogens Examples: fluorine, chlorine

- 7 valence electrons
- Form mononegative ions
- Very reactive
- React with alkali and alkaline earth metals

Group 18: Noble Gases Examples: helium, neon, argon

- 8 valence electrons
- Little tendency to establish bonds
- Low reactivity
- Odorless and colorless

11.2 Chemical Bonds

11.2.1 Covalent bond

Definition: Covalent bond occurs between non-metal atoms, where there is a sharing of electron pairs.

Octet Rule: Non-metallic elements have a tendency to establish covalent bonds to achieve the electronic distribution of the noble gas.

Types of Covalent Bonds:

- Single bonds (sharing one electron pair)
- Double bonds (sharing two electron pairs)
- Triple bonds (sharing three electron pairs)

Lewis Notation: Used to represent covalent bonds through dots representing valence electrons.

11.2.2 Ionic bond

Definition: Ionic bond occurs between metals and non-metals, and results from the attractive electric force between cations and anions.

Characteristics:

- Forms ionic networks with opposite charges
- Results in crystalline and brittle substances
- Ions organize in crystalline structures

11.2.3 Metallic bond

Definition: Metallic bond occurs between metals, where there is sharing of free electrons.

Characteristics:

- Delocalized valence electrons with free movement
- Allows electrical conductivity
- Electrons move freely through metallic structure

11.2.4 Hydrocarbons

Definition: Hydrocarbons are compounds constituted only by carbon and hydrogen.

Importance of Carbon:

- Fundamental element in organic chemistry
- Capacity to form multiple bonds
- Base of countless compounds

Types of Hydrocarbons:

1. **Saturated Compounds - Alkanes:**
   - Methane (CH)
   - Ethane (C H)
   - Propane (C H)
   - Butane (C H)
2. **Unsaturated Compounds - Alkenes:**
   - Ethene (C H)
   - Propene (C H)
   - Butene (C H)
3. **Unsaturated Compounds - Alkynes:**
   - Ethyne (C H)
   - Propyne (C H)
   - Butyne (C H)

Sources of Hydrocarbons:

- Coal
- Oil
- Natural gas

Applications:

- Production of plastics
- Production of fuels
- Chemical industry

Carbon Allotropes:

- Diamond
- Graphite
- Graphene

(Each allotrope has distinct properties due to different organization of carbon atoms)

Content adapted from O Bichinho do Saber (www.obichinhodosaber.com)